

Vivyn Bhavani Raaman

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EDUCATION

University of Connecticut, Storrs, CT (3.9 GPA) Expected May 2028
Bachelor of Science in Mechanical Engineering, Aerospace Concentration | Junior Standing (Fall 2026)

Central Connecticut State University, New Britain, CT August 2024 – May 2025

TECHNICAL SKILLS

CAD & Design: SolidWorks, Siemens NX, Autodesk Inventor, Form/Fit/Function (FFF) Analysis, Component Selection, Change Documentation

Simulation & CFD: ANSYS Fluent, OpenFOAM, ParaView, PyVista, Airfoil Analysis, Angle-of-Attack Sweep Studies, Aerodynamics

Manufacturing & Testing: Manual Lathe, Drill Press, Intro CNC Machining, Wind Tunnel Test Part Design, COTS Sourcing (McMaster-Carr)

Documentation: Technical Documentation, Excel, PowerPoint, Word, Adobe Acrobat

ENGINEERING EXPERIENCE

UConn Formula SAE, Storrs, CT September 2025 – Present
Aerodynamics & Composites Subsystem

- Identified in CFD that CT18 front-wing upwash reduced rear-wing downforce below its isolated performance; designed an inverted wing to redirect upwash away from the rear wing with minimal drag.
- Ran airfoil sweeps across four airfoils and multiple angles of attack (AoA), built the 3D CAD, and simulated candidates with the full car in ANSYS Fluent, increasing the car's downforce by 8%.
- Currently designing sidepod and validating radiator setup as part of the aerodynamics subsystem.
- Designed parts for wind tunnel testing and collaborated with cross-functional engineering subteams.

Central Connecticut State University, New Britain, CT February 2025 – May 2025
Research Fellow, Senior Vertical Axis Wind Turbine (VAWT) Project

- Supported senior VAWT design projects by reviewing design documentation and evaluating mechanical components.
- Selected and compared components using McMaster-Carr and authored documentation justifying each choice on form, fit, and function.
- Prepared prototype evaluation and testing documentation.

WORK EXPERIENCE

Friendly's, Avon, CT March 2023 – Present
Fountain / Cashier

- Coordinate with coworkers to manage customer flow and fulfill orders on time in a fast-paced restaurant environment.
- Handle register transactions and customer requests accurately while balancing teamwork and time management.

ACTIVITIES

American Society of Mechanical Engineers (ASME), CCSU August 2024 – May 2025

- Member; took part in engineering workshops, team-based technical projects, and competitions.